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(54) **SYSTEMS AND METHODS FOR MULTIPLE REPEATER AGGREGATION**

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(57) **ABSTRACT**

Related U.S. Application Data

(60) Provisional application No. 63/279,581, filed on Nov. 15, 2021.

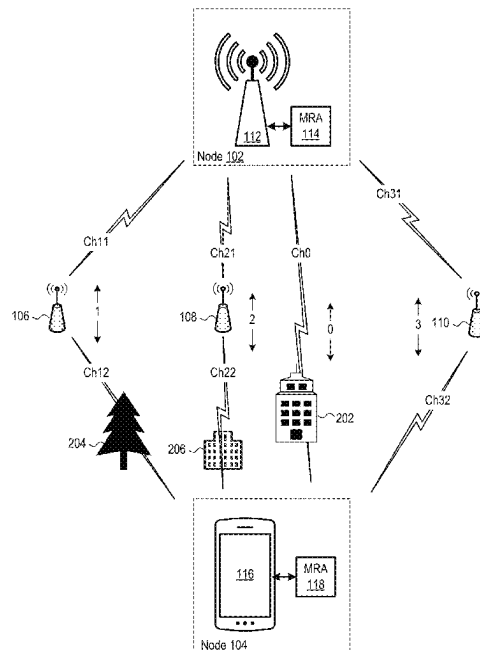
(51) **Int. Cl.**
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(52) **U.S. Cl.**
CPC **H04W 40/12** (2013.01); **H04B 7/155**
(2013.01)

(58) **Field of Classification Search**
CPC H04W 40/12; H04B 7/155
See application file for complete search history.

A method for multiple repeater aggregation includes (a) receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, where each of the wireless communication paths includes a respective radio frequency (RF) repeater, and (b) selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths.

17 Claims, 7 Drawing Sheets



100

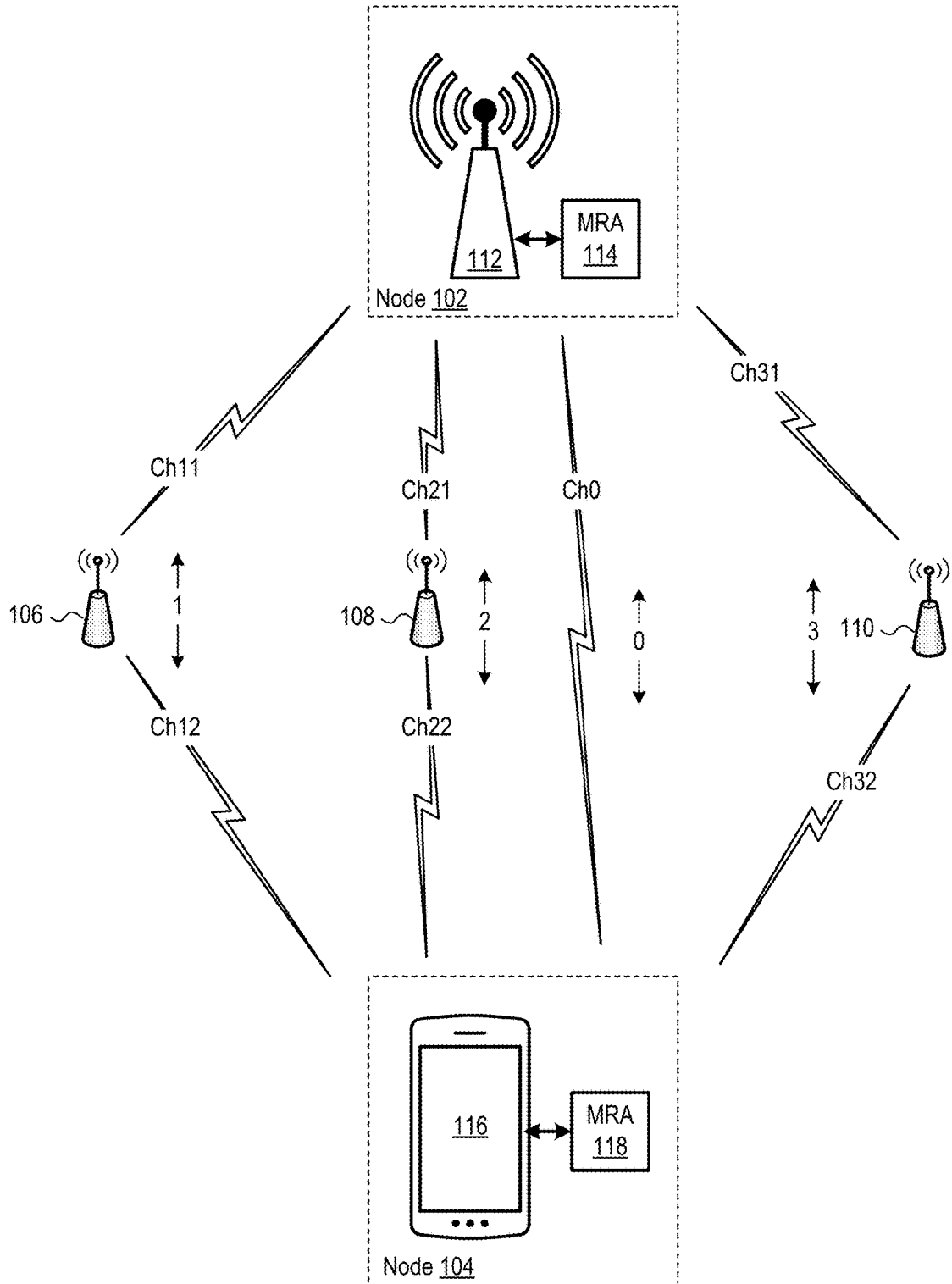


FIG. 1

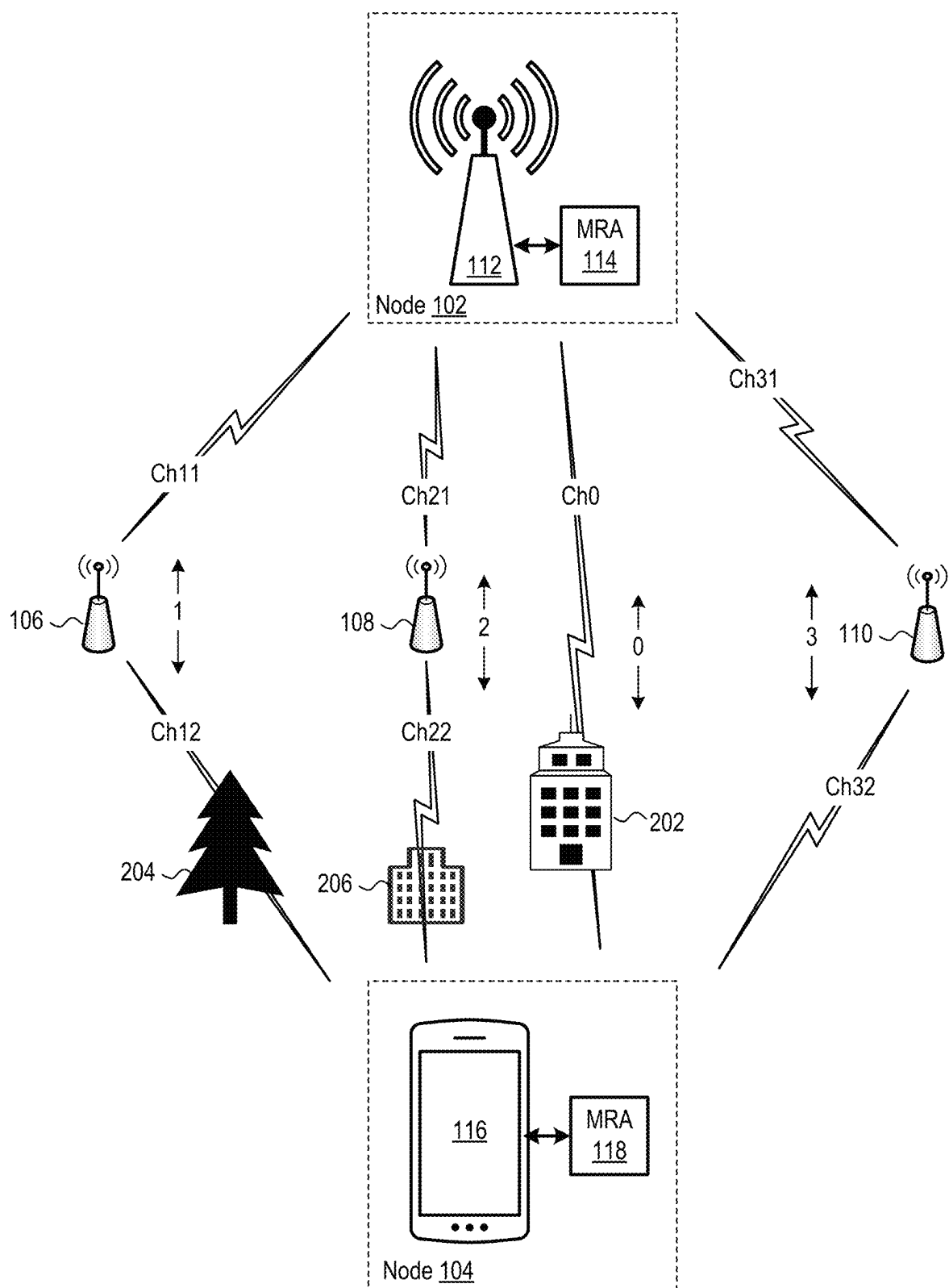


FIG. 2

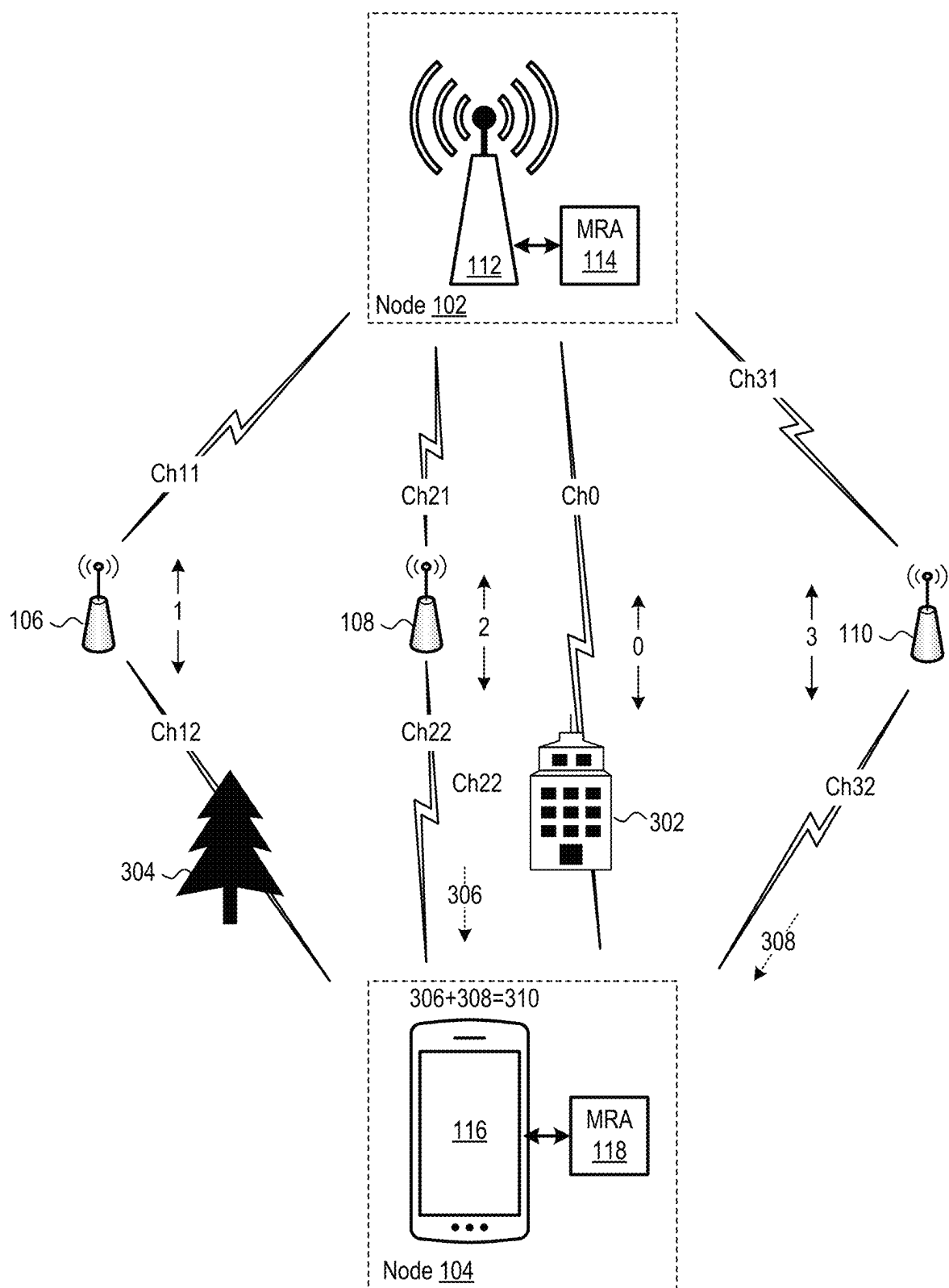


FIG. 3

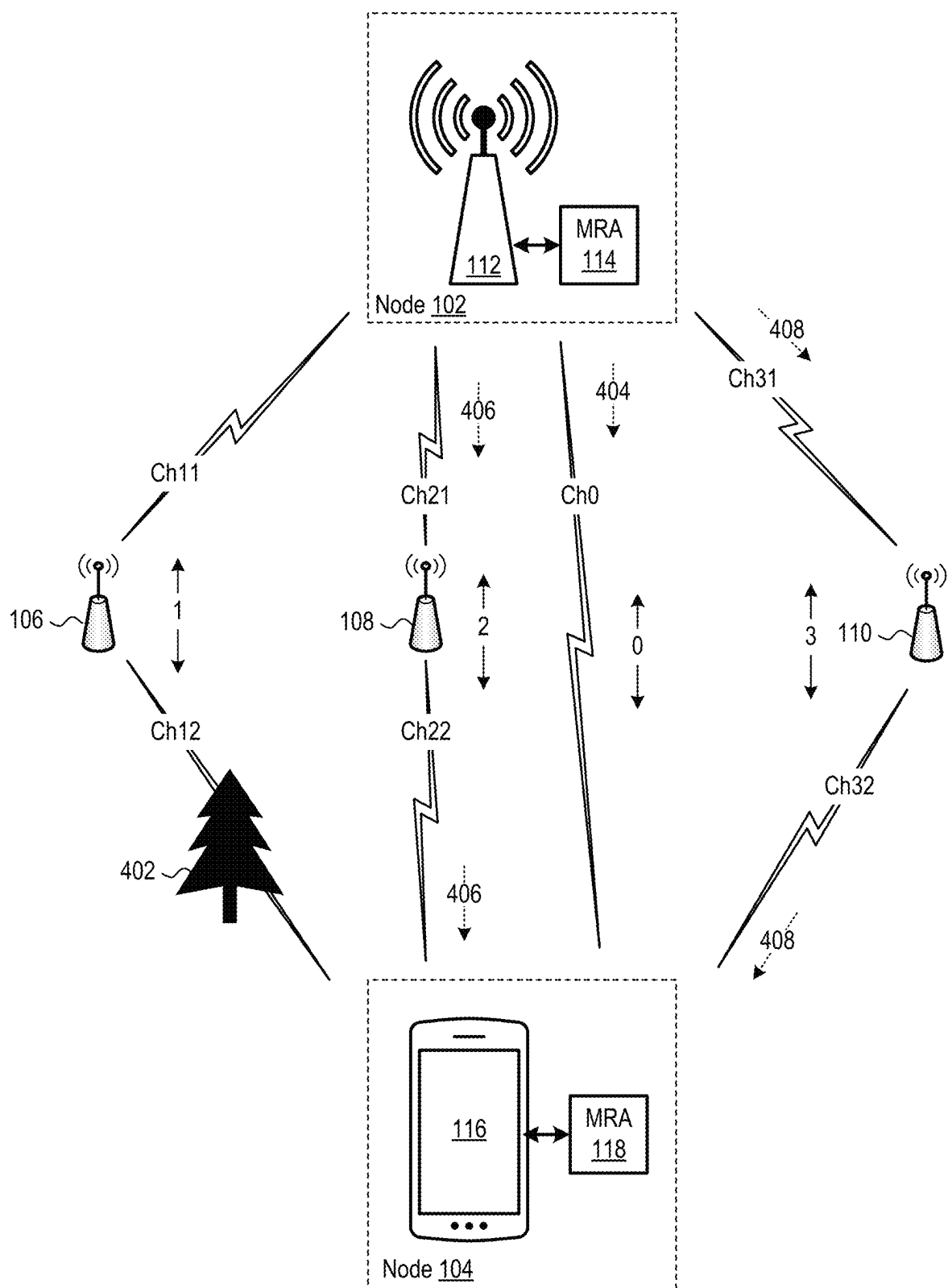


FIG. 4

500

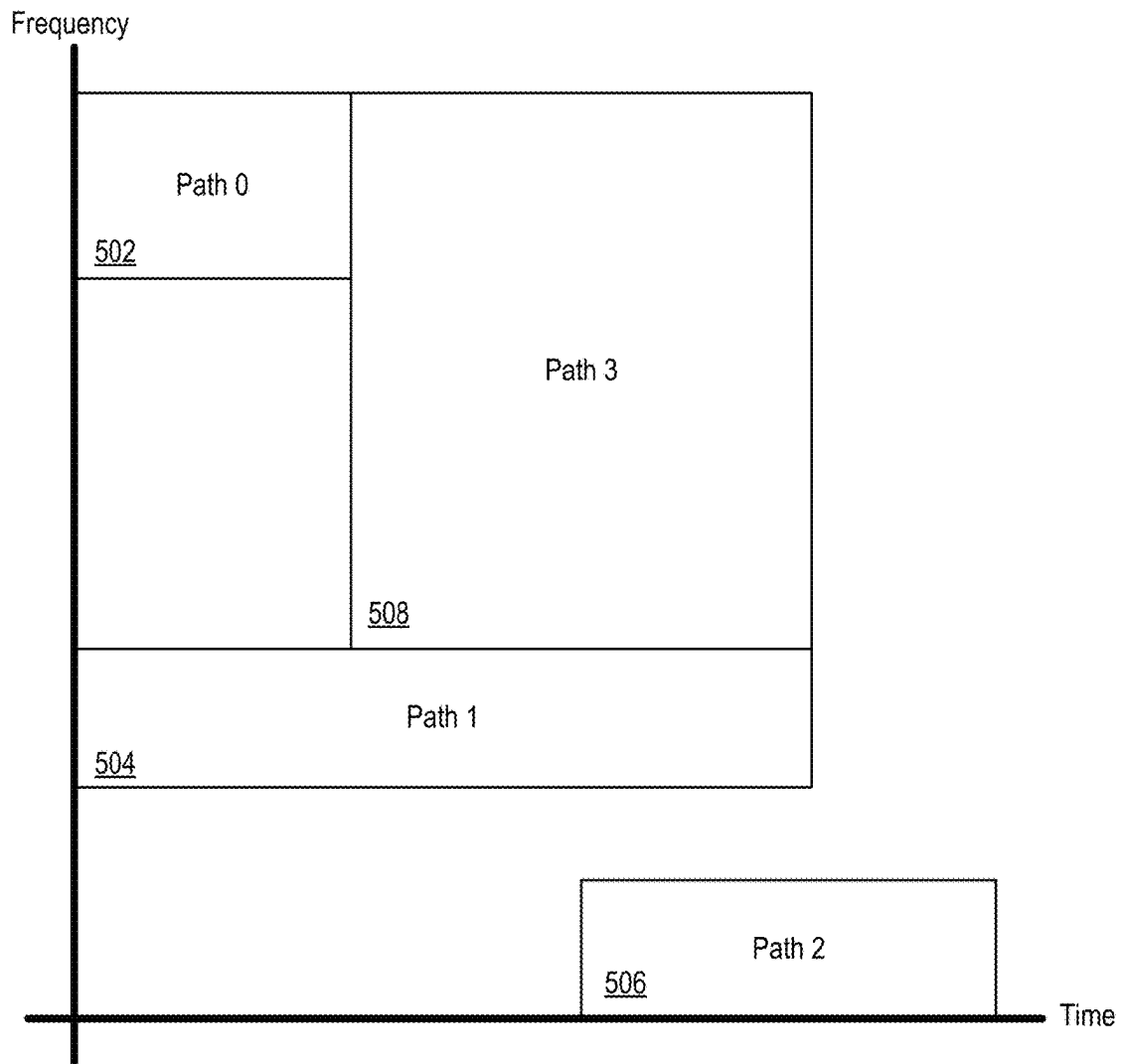


FIG. 5

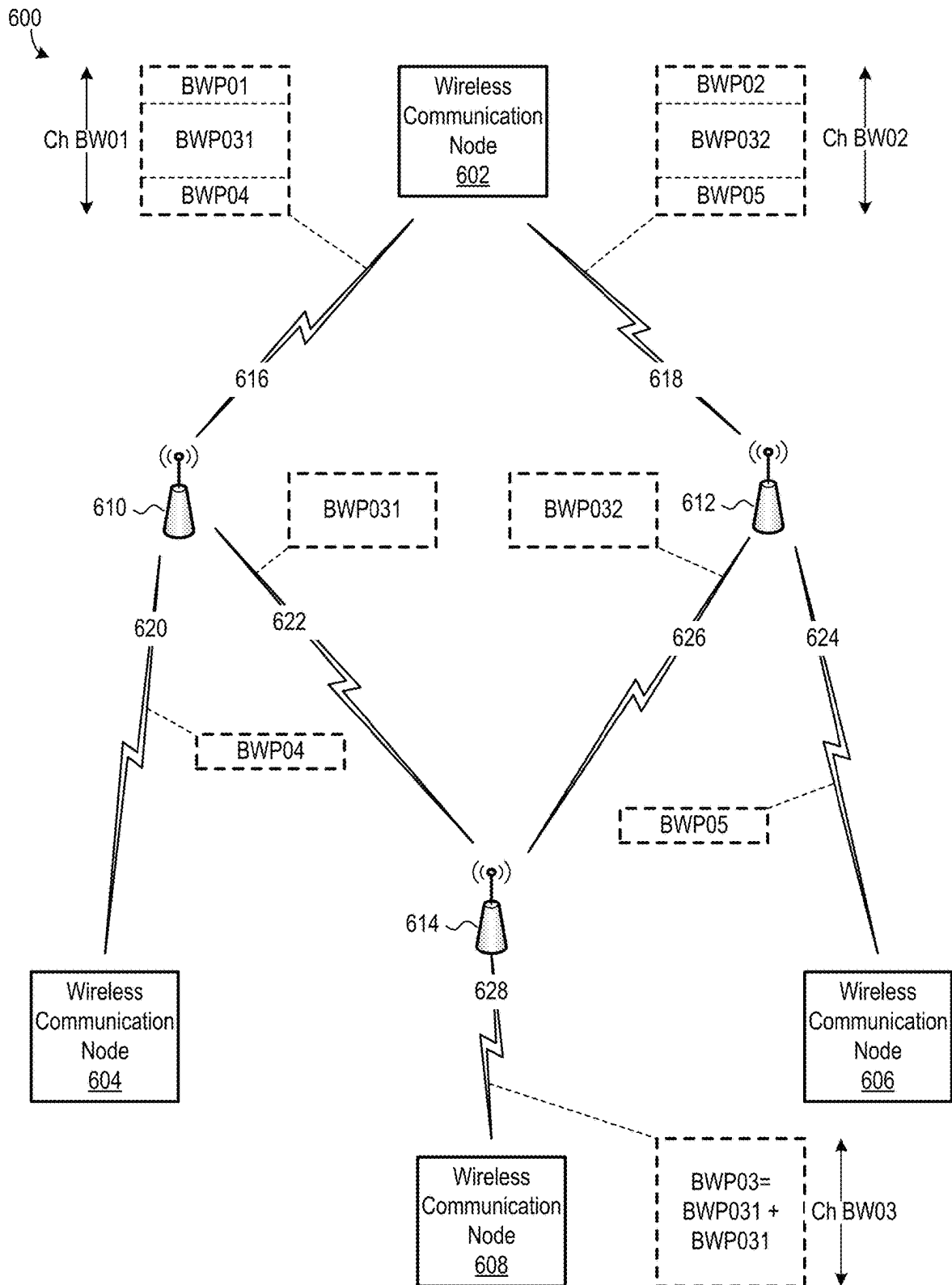


FIG. 6

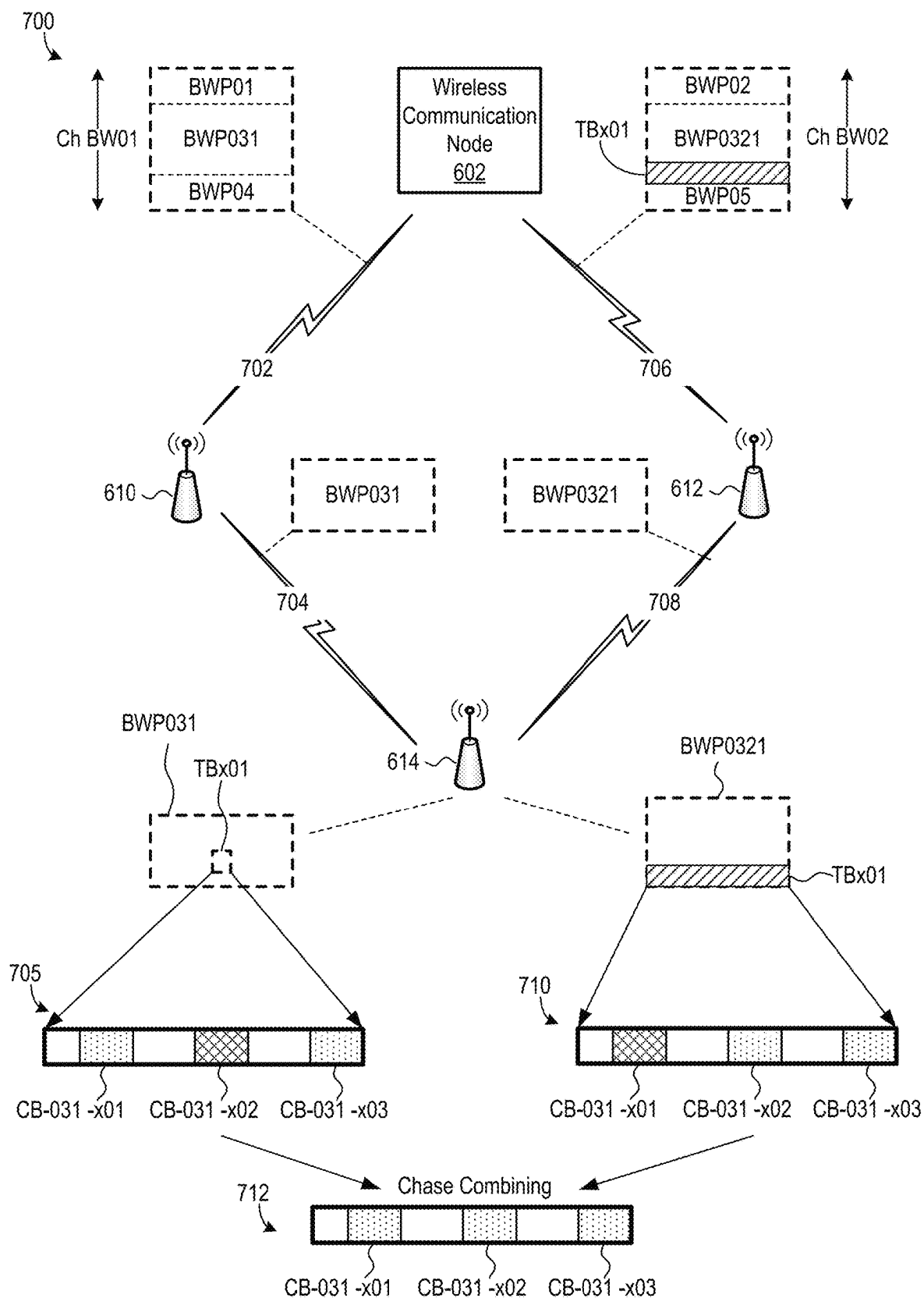


FIG. 7

SYSTEMS AND METHODS FOR MULTIPLE REPEATER AGGREGATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/279,581, filed on Nov. 15, 2021, which is incorporated herein by reference.

BACKGROUND

Wireless communication networks transmit wireless communication signals between two or more wireless communication nodes. Examples of wireless communication nodes include, but are not limited to, wireless base stations, user equipment (UE) devices, and radio frequency (RF) repeaters. As such, performance of a wireless communication network is at least partially a function of quality of wireless communication links between wireless communication nodes.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a wireless communication network including a system for multiple repeater aggregation (MRA), according to an embodiment.

FIG. 2 is a block diagram illustrating an embodiment of a method for MRA including selecting a single wireless communication path to transmit data between two wireless communication nodes.

FIG. 3 is a block diagram illustrating an embodiment of a method for MRA including combining copies of a common wireless communication signal.

FIG. 4 is a block diagram illustrating an embodiment of a method for MRA including joint transmission.

FIG. 5 is a graph illustrating one example allocation of communication resources in frequency and time across wireless communication paths of the FIG. 1 wireless communication network.

FIG. 6 is a block diagram of a wireless communication network illustrating an example of a method for load balancing, according to an embodiment.

FIG. 7 is a block diagram of an embodiment of the FIG. 6 wireless communication network configured to retransmit a data structure via an alternate path, in response to the data structure including errors.

DETAILED DESCRIPTION OF THE EMBODIMENTS

A distance between two wireless communication nodes may be too long to transmit robust wireless communication signals between the two wireless communication nodes. Additionally, an obstacle, such as by a tree, a building, or a natural feature, may block direct transmission of wireless communication signals between two wireless communication nodes. Radio frequency (RF) repeaters, and similar devices, may be deployed to at least partially overcome such problems by extending wireless communication network range and/or by providing alternate paths for wireless communication signals to travel between wireless communication nodes.

A standard RF repeater, sometimes referred to as a RF relay, simply receives wireless communication signals, amplifies the wireless communication signals, and forwards or “repeats” the wireless communication signals to another

wireless communication node. For example, the Third Generation Partnership Project (3GPP) specifies a RF repeater in the fifth generation (5G) new radio (NR) Release 17 as a device which works on a RF layer and simply amplifies and forwards all wireless communication signals it receives. A standard RF repeater does not have wireless communication beam management capability, and a standard RF repeater is typically omnidirectional. While a standard RF repeater supports full-duplex communication, a standard RF repeater is not capable of distinguishing between downlink data and uplink data. Additionally, a standard RF repeater amplifies noise and interference along with wireless communication signals. As such, deployment of a standard RF repeater in a wireless communication network may or may not improve signal to interference plus noise ratio (SINR) in the wireless communication network, depending on the environment and the operating conditions of the wireless communication network.

A smart RF repeater (SR) has more functionality than a standard RF repeater. For example, the 3GPP 5G NR Release 18 specifies a smart RF repeater as a device which works on the RF layer of the user plane, and well as on the physical (PHY) and RF layers of the control plane. Release 18 also specifies that a smart repeater supports wireless communication beam management and that a smart repeater is capable of distinguishing between downlink data and uplink data.

Disclosed herein are new systems and methods which advance the state of the art of mesh wireless communication networks. Certain embodiments are operable with essentially any device which repeats wireless communication signals, including but not limited to, RF repeaters, RF relays, smart RF repeaters, and Integrated Access Backhaul (IAB) devices. Other embodiments leverage functionality of smart RF repeaters and similar devices.

The new systems and methods can be implemented in essentially any type of wireless communication network. For example, the new systems and methods could be embodied in 3GPP wireless communication networks, including, but not limited to, long term evolution (LTE) wireless communication networks, fifth generation (5G) wireless communication networks, sixth generation (6G) wireless communication networks, and future developed 3GPP wireless communication networks. As another example, the new systems and methods could be embodied in non-3GPP wireless communication networks, including, but not limited to, Wi-Fi wireless communication networks, satellite wireless communication networks (e.g., using very low earth orbit (VLEO) satellites, low earth orbit (LEO) satellites, medium earth orbit (MEO) satellites, or geostationary equatorial orbit (GEO) satellites), Bluetooth wireless communication networks, long range (LoRa) wireless communication protocol, Zigbee wireless communication networks, Z-Wave wireless communication networks, etc. Furthermore, the new systems and methods could be embodied in wireless communication networks using licensed RF spectrum, unlicensed RF spectrum, or a combination of licensed and unlicensed RF spectrum.

Multiple Repeater Aggregation

Applicant has developed systems and methods for multiple repeater aggregation (MRA) which leverage multiple devices which repeat wireless communication signals, such as in a mesh wireless communication network. Certain embodiments are capable of improving wireless communication network coverage and reliability, increasing wireless communication channel capacity for downlink data and/or uplink data, and or extending wireless communication net-

work coverage. For example, some embodiments enable a wireless communication network to maintain service to a wireless communication node even if some wireless communication signals to the wireless communication node are blocked by an obstacle, e.g., one wireless communication beam is blocked, thereby helping prevent shadowing loss. As another example, certain embodiments benefit from diversity gain, which mitigates small scale fading. As an additional example, particular embodiments are capable of serving a given wireless communication node via multiple channels with low correlation between the channels, or even with no correlation between the channels, which increases channel capacity according to Shannon theory.

While the new systems and methods for MRA may achieve advantages in essentially any application, they may be especially beneficial in fixed wireless access (FWA) applications, vehicular applications, and aircraft applications. In particular, FWA applications may be particularly challenging due to FWA customer premises equipment (CPE) frequently being located indoors, as well as due to applications typically requiring more throughput than mobile applications. Use of the new MRA systems and methods may help improve coverage indoors and help achieve high throughput, which may make them particularly beneficial in FWA applications. Vehicular applications may also be particularly challenging due to vehicle movement, such as due to doppler shift resulting from vehicle movement. Use of the new MRA systems and methods may help mitigate such challenges. Furthermore, aircraft, such as drones, may be difficult to adequately serve using conventional wireless communication networks because conventional wireless communication networks are typically optimized for terrestrial users with down tilt in wireless base station antennas, causing coverage towards the sky to be poor. The new MRA systems and methods may be used in wireless communication networks to improve coverage towards the sky, which benefits aircraft applications.

FIG. 1 is a block diagram of a wireless communication network 100 which includes an embodiment of the new systems for MRA. Wireless communication network 100 includes a wireless communication node 102, a wireless communication node 104, a RF repeater 106, a RF repeater 108, and a RF repeater 110. It is understood, however, that wireless communication network 100 can include any quantity of wireless communication nodes and any quantity of RF repeaters, as long as wireless communication network 100 includes at least two wireless communication nodes and at least two RF repeaters. Particular embodiment of the new systems and methods for MRA do not require coordination among devices which repeat wireless communication signals. As such, in certain embodiments, RF repeaters 106, 108, and 110 are conventional RF repeaters or conventional RF relays. However, RF repeaters 106, 108, and 110 could be more sophisticated devices, such as smart RF relays or IAB devices, as well as custom devices, without departing from the scope hereof.

Each of wireless communication node 102 and wireless communication node 104 is configured to transmit and receive wireless communication signals. Wireless communication node 102 includes a wireless base station 112 and a MRA module 114. In some embodiments, wireless base station 112 includes a 3GPP wireless base station, including, but not limited to, a NodeB, an eNode B, a gNodeB, or a future-developed 3GPP wireless base station. In some other embodiments, wireless base station 112 is a non-3GPP wireless base station, including, but not limited to, a Wi-Fi wireless base station, a satellite wireless base station, a

Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc. It should be noted that wireless base station 112 need not be a complete wireless base station. For example, in particular embodiments, wireless base station 112 is a remote radio head of a wireless base station. Additionally, wireless base station 112 could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a user equipment (UE) device.

Wireless communication node 104 includes a UE device 116 and a MRA module 118. Although UE device 116 is depicted as being a mobile phone, UE device 106 could be another type of UE device. For example, UE device 106 could alternately be a computer, a set-top device, a data storage device, an Internet of Things (IoT) device, an entertainment device, a computer networking device, a modem, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, or a wireless access device (including, for example, an eNB, a gNB, a Wi-Fi-based wireless access point, an IAB access point, a microcell, a picocell, a femtocell, a macrocell, a Wi-Fi-based application, a satellite communication device, etc.). Additionally, UE device 116 could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a wireless base station.

Wireless communication network 100 includes four wireless communication paths, i.e., wireless communication path 0, wireless communication path 1, wireless communication path 2, and wireless communication path 3, between wireless communication nodes 102 and 104. Wireless communication path 0 is a direct path between wireless communication node 102 and wireless communication node 104 via a wireless channel Ch0. Accordingly, wireless communication path 0 does not include a RF repeater. Each of wireless communication paths 1, 2, and 3 includes a RF repeater. Specifically, wireless communication path 1 includes (i) a wireless channel Ch11 between wireless communication node 102 and RF repeater 106 and (ii) a wireless channel Ch12 between RF repeater 106 and wireless communication node 104. Wireless communication path 2 includes (i) a wireless channel Ch21 between wireless communication node 102 and RF repeater 108 and (ii) a wireless channel Ch22 between RF repeater 108 and wireless communication node 104. Wireless communication path 3 includes (i) a wireless channel Ch31 between wireless communication node 102 and RF repeater 110 and (ii) a wireless channel Ch32 between RF repeater 110 and wireless communication node 104. Any of wireless communication paths 0, 1, 2, and 3 could include a plurality of wireless communication beams, and each wireless communication beam could include a plurality of streams. Any one of wireless communication paths 0, 1, 2, and 3 could carry downlink data, uplink data, or a combination of downlink data and uplink data.

It is understood that the number of wireless communication paths between wireless communication nodes 102 and 104 will vary with the number of RF repeaters in wireless communication network 100. Additionally, in some embodiments, one or more wireless communication paths between wireless communication nodes 102 and 104 may span two or more RF repeaters. Furthermore, a given wireless communication path between wireless communication nodes 102 and 104 may not be available due to, for example, the path being blocked by an obstacle.

MRA module 114 and MRA module 118 collectively form an embodiment of the new systems for MRA. In some alternate embodiments, however, one of MRA module 114 and MRA module 118 is omitted, such that solely MRA module 114 or solely MRA module 118 forms the system for MRA. Each of MRA modules 114 and 118 is implemented, for example, by a processor (not shown) executing instructions in the form of software and/or firmware stored in a data store (not shown). In some embodiment, MRA modules 114 and 118 are at least partially implemented by electronic circuitry (not shown). While MRA module 114 is depicted as being separate from wireless base station 112, MRA module 114 may be partially or fully integrated with wireless base station 112. For example, in certain embodiments, MRA module 114 is implemented by a processor of wireless base station 112 executing instructions in the form of software and/or firmware stored in a data store (e.g., a memory or a hard drive) of wireless base station 112. Similarly, although MRA module 118 is depicted as being separate from UE device 116, MRA module 118 may be partially or fully integrated with UE device 116. For example, in certain embodiments, MRA module 118 is implemented by a processor of UE device 116 executing instructions in the form of software and/or firmware stored in a data store, e.g., a memory, of UE device 116.

The system for MRA of wireless communication network 100 is configured to execute one or more of the following methods for MRA, to promote high performance of wireless communication network 100: (a) Method A—select of a single path, (b) Method B—combine copies of a signal, (c) Method C—joint transmission, (d) Method D—dynamic beam configuration, (e) Method E—beam direction, (f) Method F—multiple streams, (g) Method G—frequency/time resource aggregation, (h) Method H—fake base station detection, and (i) Method I—dynamic power control.

Method A—Select a Single Path

In method A, each RF repeater 106, 108, and 110 amplifies and repeats a received downlink or uplink wireless communication signal. Additionally, each RF repeater 106, 108, and 110 may receive the same copy of a downlink wireless communication signal or an uplink wireless communication signal sent by wireless base station 112 or UE device 116, respectively. As such, there may be multiple wireless communication paths for the same copy of a wireless communication signal between wireless communication node 102 and wireless communication node 104. However, these multiple wireless communication paths may not be equal. For example, one RF repeater 106, 108, and 110 may be blocked while another RF repeater 106, 108, and 110 may not be blocked. In method A, MRA module 114 and/or MRA module 118 select a single wireless communication path to transmit data between wireless communication node 102 and wireless communication node 104, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes.

In particular, one or more devices of wireless communication network 100 measure conditions of one or more wireless communication paths, or portions of wireless communication paths, of wireless communication network 100, and the devices send data representing the measured conditions to MRA module 114 and/or MRA module 118. For example, wireless base station 112 may measure conditions of wireless channels Ch0, Ch11, Ch21, and Ch31 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. As another example, UE device 116 may measure conditions of wireless channels

Ch0, Ch12, Ch22, and Ch32 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. As additional example, RF repeater 106 may measure conditions of wireless channels Ch11 and Ch12 and send data representing the measured conditions to MRA module 114 and/or MRA module 118. Examples of possible measured conditions of wireless communication paths include, but are not limited to, one or more of reference signal received power (RSRP) of each wireless communication path, reference signal received quality (RSRQ) of each wireless communication path, time advance of each wireless communication path, cell ID of each wireless communication path, wireless communication beam ID of each wireless communication path, percentage of utilization of each wireless communication path, etc.

MRA module 114, MRA module 118, or the combination of MRA module 114 and MRA module 118, select one wireless communication path to transmit data between wireless communication node 102 and wireless communication node 104 at least partially based on the data representing the measured conditions. For example, MRA module 114 and/or MRA module 118 may select whichever wireless communication path 1, 2, 3, or 4 which has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module 114 and/or MRA module 118 then cause data to be routed between wireless communication node 102 and wireless communication node 104 solely via the one selected wireless communication path, such as by controlling wireless base station 112 and/or UE device 116 to route data accordingly. It should be noted that particular embodiments of Method A do not require coordination of RF repeater 106, RF repeater 108, or RF repeater 110, thereby enabling the RF repeaters to be conventional RF repeaters in Method A.

FIG. 2 is a block diagram of wireless communication network 100 illustrating one example of Method A. In this example, wireless communication paths 0, 1, and 2 are partially or completely blocked by a building 202, a tree 204, and a building 206, respectively. MRA module 114 and/or MRA module 118 may analyze one or more measured conditions of wireless paths 0, 1, 2, and 3 and determine that path 3 is the best path to transfer data between wireless communication node 102 and wireless communication node 104, such as in response to wireless communication path 3 having a highest RSRP, a highest RSRQ, a smallest time advance, or a smallest percentage of wireless communication path. In response thereto, MRA module 114 and/or MRA module 118 cause wireless base station 112 and/or UE device 116 to route data between 102 and wireless communication node 104 solely via wireless communication path 1, i.e., using channel Ch31, RF repeater 110, and Ch32.

Method B—Combine Copies of a Signal

Referring again to FIG. 1, in method B, multiple copies of a wireless communication signal, i.e., either a downlink wireless communication signal or an uplink wireless communication signal, are transmitted between wireless base station 112 and UE device 116 via different respective wireless communication paths. In contrast with Method A, multiple copies of a wireless communication signal may be used by wireless base station 112 and UE device 116 in Method B. In particular, MRA module 114 and/or MRA module 118 cause wireless base station 112 to combine copies of an uplink wireless communication signal it receives via two or more of wireless communication paths 0, 1, 2, and 3 to generate a combined wireless communication signal. Similarly, MRA module 114 and/or MRA module 118 cause UE device 116 to combine copies of a downlink

wireless communication signal it receives via two or more of wireless communication paths **0**, **1**, **2**, and **3** to generate a combined wireless communication signal. Each combined wireless communication signal may have a greater amplitude than any one of its constituent copies of the wireless signal communication signal, which advantageously helps achieve a high signal to noise ratio. In some embodiments, the copies of the wireless communication signal are combined using a soft combining process. For example, wireless base station **112** may sum complex values of received copies of an uplink wireless communication signal to generate a combined uplink wireless communication signal, and UE device **116** may sum complex values of received copies of a downlink wireless communication signal to generate a combined downlink wireless communication signal. Certain embodiments of Method B do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters in Method B.

Some embodiments of MRA module **114** and/or MRA module **118** are configured to cause wireless base station **112** and UE device **116** to combine only a predetermined quantity of received copies of a wireless communication signal, or to combine only received copies of a wireless communication signal that meet one or more predetermined criteria. For example, in one embodiment, MRA module **114** and/or MRA module **118** are configured to cause wireless base station **112** or UE device **116** to combine the two received copies of a wireless communications signal having largest RSRP values, largest RSRQ values, or smallest time advance values. As another example, in another embodiment, MRA module **114** and/or MRA module **118** are configured to cause wireless base station **112** or UE device **116** to combine only copies of a wireless communication signal having RSRP values exceeding a threshold value, RSRQ values exceeding a threshold value, or time advance values that are below a threshold value.

FIG. 3 is a block diagram of wireless communication network **100** illustrating one example of Method B. In this example, wireless communication paths **0** and **1** are blocked by a building **302** and a tree **304**, respectively. Consequently, wireless communication node **104** is capable of receiving copies of a downlink wireless communication signal solely via wireless communication paths **2** and **3**. Accordingly, UE device **116** receives copies **306** and **308** of a common downlink wireless communication signal, and UE device **116** combines these two copies to generate a combined downlink wireless communication signal **310**, as symbolically shown by text “**306+308=310**” in wireless communication node **104**. The magnitude of combined downlink wireless communication signal **310** is greater than the respective magnitudes of either of copies **306** and **308** of the downlink wireless communication signal. Therefore, Method B increases signal to noise ratio of downlink wireless communication signals processed by UE device **116**. UE device **116** could analogously receive and combine copies of an uplink wireless communication sent via wireless communication paths **2** and **3**, in the FIG. 3 example. Method C—Joint Transmission

Referring again to FIG. 1, in Method C, MRA module **114** and MRA module **118** cooperate such that data is transferred between wireless communication node **102** and wireless communication node **104** using joint transmission, such that different downlink data and/or uplink data are transferred via two or more of wireless communication paths **0**, **1**, **2**, and **3**. For example, in particular embodiments, MRA modules **114** and **118** are configured such that wireless base station **112**

sends different downlink data to each of RF repeaters **106**, **108**, and **110** via respective wireless communication signals in wireless channels Ch**11**, Ch**21**, and Ch**31**. RF repeaters **106**, **108**, and **110** then forward the received downlink data to UE device **116** via respective wireless communication signals in each of wireless channels Ch**12**, Ch**22**, and Ch**32**, and UE device **116** aggregates the received wireless communication signals. Accordingly, UE device **116** receives downlink data from wireless base station **112** via multiple wireless communication paths. Similarly, in certain embodiments, MRA modules **114** and **118** are configured such that UE device **116** sends different uplink data to each of RF repeaters **106**, **108**, and **110** via respective wireless communication signals in each of wireless channels Ch**12**, Ch**22**, and Ch**32**. RF repeaters **106**, **108**, and **110** then forward the received uplink data to wireless base station **112** via respective wireless communication signals in each of wireless channels Ch**11**, Ch**21**, and Ch**31**, and wireless base station **112** aggregates the received wireless communication signals. Accordingly, wireless base station **112** receives different uplink from UE device **116** via multiple wireless communication paths. Direct path **0**, i.e., wireless channel Ch**0**, may be additionally be used in conjunction with one or more of wire communication paths **1**, **2**, and **3** to further increase number of data transfer paths between wireless communication nodes **102** and **104**.

Such ability to send different data via multiple wireless communication paths in Method C enables wireless communication network **100** to achieve a significantly larger throughput than would be feasible using conventional techniques. Additionally, the multiple wireless communication paths will typically be at different angles with respect to each of wireless base station **112** and UE device **116**, such that there may be small correlation, or even essentially no correlation between wireless communication paths **0**, **1**, **2**, and **3**, which further increases maximum theoretical throughput of wireless communication network **100**. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network **100** helps mitigate shadowing, small-scale fading, and Doppler shift. Furthermore, certain embodiments of Method C do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters.

FIG. 4 is a block diagram of wireless communication network **100** illustrating one example of Method C. In this example, wireless communication path **1** is blocked by a tree **402**. However, wireless communication paths **0**, **2**, and **3** are available for data transmission between wireless communication nodes **102** and **104**. Accordingly, MRA module **114** and MRA module **118** cooperate to provide for joint transmission of data between wireless base station **112** and UE device **116** using wireless communication paths **0**, **2**, and **3**. For example MRA module **114** and MRA module **118** may cooperate such that wireless base station **112** sends different downlink wireless communication signals **404**, **406**, and **408** to UE device **116** in wireless communication via paths **0**, **2**, and **3**. Specifically, in this example, wireless base station **112** sends downlink wireless communication signals **404** directly to UE device **116** using wireless communication path **0**, and wireless base station **112** sends downlink wireless communication signals **406** and **408** via wireless channels Ch**21** and Ch**31** to RF repeater **108** and RF repeater **110**, respectively. RF repeaters **108** and **110** then forward downlink wireless communication signals **406** and **408** to UE device **116**, respectively, and UE device **116** aggregates received wireless communication signals **404**, **406**, and **408**.

Method D—Dynamic Beam Configuration

Referring again to FIG. 1, in certain embodiments, each wireless each RF repeater **106**, **108**, and **110** may receive and transmit a plurality of beams. For example, wireless channel Ch**11** between wireless base station **112** and RF repeater **106** may include a plurality of beams, and wireless channel Ch**12** between RF repeater **106** and UE device **116** may also include a plurality of beams. As such, each of wireless communication paths **0**, **1**, **2**, and **3** may include a plurality of possible beams by which wireless communication signals may be transmitted. In method D, MRA module **114** and/or MRA module **118** select a single wireless communication beam, or a single set of wireless communication beams, of a single wireless communication path **0**, **1**, **2**, or **3** to transmit data between wireless communication node **102** and wireless communication node **104**, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes. As such, method D is similar to method A, but method D further includes wireless communication beam coordination. The measured conditions of wireless communication paths in method D may include measured conditions of respective wireless communication beams of the wireless communication paths.

In certain embodiments of Method D, UE device **116** may measure conditions of each wireless communication beam received from each RF repeater **106**, **108**, and **110**, and UE device **116** may send data representing the measured conditions of the wireless communication beams to MRA module **114** and/or MRA module **118**. Similarly, wireless base station **112** may measure conditions of each wireless communication beam received from each RF repeater **106**, **108**, and **110**, and wireless base station **112** may send data representing the measured conditions of the wireless communication beams to MRA module **114** and/or MRA module **118**. Such measurements are performed, for example, on a periodic basis, on an aperiodic basis, or in response to event. MRA module **114** and/or MRA module **118** may then select a single wireless communication beam, or a single set of wireless communication beams, for transmitting data between wireless communication nodes **102** and **104**, based at least in part on the data representing the measured conditions of the wireless communication beams. For example, MRA module **114** and/or MRA module **118** may select whichever wireless communication beam, or set of wireless communication beams, has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module **114** and/or MRA module **118** then cause data to be routed between wireless communication node **102** and wireless communication node **104** solely via the one selected wireless communication beam, or selected set of wireless communication beams, such as by controlling wireless base station **112** and/or UE device **116** to route data accordingly.

In particular embodiments, wireless base station **112** and/or UE device **116** may not distinguish between wireless communication beams sent directly between the two devices and wireless communication beams sent between the two devices via one or more RF repeaters **106**, **108**, and **110**. Additionally, some embodiments of Method D do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters.

Method E—Beam Direction

Method E is an alternate embodiment of Method B further including wireless communication beam coordination. In Method E, multiple copies of a downlink wireless commu-

nication signal or an uplink wireless communication signal are sent by wireless base station **112** or UE device **116** via different respective wireless communication beams. For example, wireless base station **112** may send four copies of a common downlink wireless communication signal via one or more wireless communication beams along each of wireless communication paths **0**, **1**, **2**, and **3**. As another example, UE device **116** may send four copies of an uplink wireless communication signal via one or more wireless communication beams along each of wireless communication paths **0**, **1**, **2**, and **3**. Each RF repeater **106**, **108**, and **110** amplifies and forwards each wireless communication beam that it receives. As such, if all four wireless communication paths **0**, **1**, **2**, and **3** are available, UE device **116** will receive four copies of a common downlink wireless communication signal via four wireless communication beams. Similarly, if all four wireless communication paths **0**, **1**, **2**, and **3** are available, wireless base station **112** will receive four copies of a common uplink wireless communication signal via four wireless communication beams. MRA module **114** and/or MRA module **118** cause wireless base station **112** to combine copies of an uplink wireless communication signal it receives via two or more wireless communication beams. Additionally, MRA module **114** and/or MRA module **118** cause UE device **116** to combine copies of a downlink wireless communication signal it receives via two or more wireless communication beams.

Method F—Multiple Streams

In particular embodiments of wireless communication network **100**, wireless each RF repeater **106**, **108**, and **110** may receive and transmit a plurality of beams, and each beam may include a plurality of streams. Method F is similar to method C, but joint transmission is extended across multiple streams in each of two or more of wireless communication paths **0**, **1**, **2**, and **3**. For example, assume that each of wireless communication paths **0**, **1**, **2**, and **3** is available, and that each path supports two streams between wireless communication nodes **102** and **104**. In this example, there are eight routes between wireless communication nodes **102** and **104**, which significantly increases throughput of wireless communication network **100** compared to a scenario where joint transmission is not used. Additionally, in a manner analogous to that discussed above with respect to Method C, these multiple wireless communication paths enabled by Method F will typically be at different angles with respect to each of wireless base station **112** and UE device **116**, such that there may be small correlation, or even essentially no correlation between paths **0**, **1**, **2**, and **3**, which further increases maximum theoretical throughput of wireless communication network **100**. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network **100** helps mitigate shadowing, small-scale fading, and Doppler shift.

Method G—Frequency/Time Resource Aggregation

Methods A-F, discussed above, advantageously implement spatial diversity of communication resources to promote high performance of wireless communication network **100**. Any of methods A-F can be modified to further implement frequency and/or time diversity of communication resources among wireless communication paths **0**, **1**, **2**, and **3**, among wireless communication beams of a given wireless communication path, and/or among streams of a given wireless communication beam. Such implementation of frequency and/or time diversity in wireless communication network **100** may further promote high throughput and high reliability.

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FIG. 5 is a graph 500 of frequency versus time illustrating one example of how frequency and time diversity may be implemented by MRA module 114 and/or MRA module 118. In the FIG. 5 example, a respective physical resource block (PRB) 502, 504, 506, and 508 is allocated to each of wireless communication path 0, wireless communication path 1, wireless communication path 2, and wireless communication path 3. Each PRB specifies a combination of frequency and time that may be used for the path. For example, wireless base station 112, RF repeater 106, and UE devices 116 are permitted to collectively use solely the frequency and time range specified by PRB 504 to implement wireless communication path 1.

Method H—Fake Base Station Detection

Method H detects a fake wireless base station. In method H, MRA module 114 and/or MRA module 118 analyze wireless communication signals received at a node in wireless communication network 100, such as wireless communication signals received by UE device 116 or wireless communication signals received by wireless base station 112, to determine angle of incidence of the received wireless communication signals. For example, MRA module 114 and/or MRA module 118 may determine angle of incidence of wireless communication signals received by UE device 116. Reception of a plurality of wireless communication signals at substantially different angles of incidence indicates that the sending wireless base station is likely a legitimate wireless base station. In contrast, reception of a plurality of wireless communication signals at substantially common angles of incidence indicates that the sending wireless base station is likely a fake wireless base station. Accordingly, particular embodiments of MRA module 114 and/or MRA module 118 are configured to determine that a particular wireless base station is fake in response to angle of incidence of received wireless communication signals from the wireless base station differing by less than a predetermined threshold value. Additionally, some embodiment of MRA module 114 and/or MRA module 118 are configured to determine that a particular wireless base station is legitimate in response to angle of incidence of received wireless communication signals from the wireless base station differing by more than a predetermined threshold value.

Method I—Dynamic Power Control

In certain embodiments of wireless communication network 100, amplification of RF repeaters 106, 108, and 110 may be adjusted, and method I includes MRA module 114 and/or MRA module 118 adjusting amplification or “power” of one or more to RF repeaters 106, 108, and 110 to optimize or more characteristics of wireless communication network 100. Method I may be incorporated with any of the methods discussed above. For example, method I may be incorporated with method B to help further increase signal to noise ratio of a sum of copies of a downlink or uplink wireless communication signal.

Load Balancing

A conventional smart RF repeater is capable of receiving data from a donor wireless communication node via a first wireless communication beam and forwarding the data to a child using a second wireless communication beam, in a first slot. The conventional smart RF repeater may be configured such that there is a semi-static linkage between the first wireless communication beam and the second wireless communication beam, and the second wireless communication beam may be dependent on the first slot. Accordingly, a conventional smart repeater is ill-suited for handling aggregated data for one or more target wireless communication

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nodes over different wireless communication beams. For example, assume a hypothetical scenario where a conventional smart RF repeater receives the following wireless communication signals from a donor: (a) a first wireless communication beam including data 1 for forwarding to a first UE device (UE 1) and (b) a second wireless communication beam including data 2 for forwarding to a second UE device (UE 2). A conventional smart RF repeater configured with a semi-static linkage between wireless communication beams would be incapable of determining how to forward data 1 and data 2 to its intended UE device.

Applicant has developed new systems and methods for load balancing which at least partially overcome the above discussed drawbacks of conventional smart repeaters. Certain embodiments are configured to associate frequency regions, or bandwidth parts (BWPs), within a given wireless communication beam with respective destination wireless communication nodes, for forwarding data. As such, a recipient smart RF repeater, or other recipient network element, can determine where to forward data carried by a wireless communication beam at least partially based on a BWP that the data occupies in the wireless communication beam. Accordingly, particular embodiments of the new systems and methods are capable of receiving data for multiple child wireless communication nodes, e.g., UE devices, customer premises equipment (CPE), and other smart RF repeaters, and forwarding the data to the child wireless communication nodes over different respective wireless communication beams.

FIG. 6 is a block diagram of a wireless communication network 600 including an embodiment of the new systems for load balancing. Wireless communication network 600 includes a wireless communication node 602, a wireless communication node 604, a wireless communication node 606, a wireless communication node 608, a smart RF repeater 610, a smart RF repeater 612, and a smart RF repeater 614. The aforesaid elements of wireless communication network 600 as configured, for example, in a mesh configuration. The number of wireless communication nodes in wireless communication network 600, as well as the number of smart repeaters in wireless communication network 600, may vary without departing from the scope hereof.

Each wireless communication node 602, 604, 606, and 608 is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node 602, 604, 606, and 608 includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, and a monitoring device. Each wireless communication node 602, 604, 606, and 608 need not have the same configuration.

Each smart RF repeater 610, 612, and 614 is configured to receive data via wireless communication beams and forward data carried by the wireless communication beams to child wireless communication nodes according to respective BWPs associated with the data. Smart RF repeaters 610, 612, and 614 could be replaced with other devices capable of forwarding wireless communication beams, e.g., IABs, as long as the other devices are capable of forwarding data carried by wireless communication beams to child nodes

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according to respective BWPs associated with the data. Additionally, one or more of smart repeaters **610**, **612**, and **614** could be combined with another device, such as a wireless base station, CPE, or a UE device.

FIG. 6 depicts an example of operation of wireless communication network **600** in load balancing where data is being transmitted from wireless communication node **602** to other nodes of wireless communication network **600**, such that wireless communication node **602** is acting as a “donor” wireless communication node. It is understood, though, that wireless communication network **600** could operate with other nodes being donor wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. 6 example.

In the FIG. 6 example, wireless communication node **602** transmits wireless communication beams **616** and **618** to smart RF repeaters **610** and **612**, respectively, such as in different respective time slots. Each of wireless communication beams **616** and **618** is structured, in the frequency domain, into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth of the beam. In particular, wireless communication beam **616** is structured into bandwidth parts BWP01, BWP031, and BWP04, which collectively encompass a bandwidth Ch BW01 of a wireless communication channel associated with wireless communication beam **616**, as illustrated in FIG. 6. Each of bandwidth parts BWP01, BWP031, and BWP04 includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part BWP01 includes data associated with smart RF repeater **610**, bandwidth part BWP031 includes data associated with smart RF repeater **614**, and bandwidth part BWP04 includes data associated with wireless communication node **604**. Wireless communication beam **618**, in turn, is structured into bandwidth parts BWP02, BWP032, and BWP05, which collectively encompass a bandwidth Ch BW02 of a wireless communication channel associated with wireless communication beam **618**, as illustrated in FIG. 6. While not required, in certain embodiments, bandwidth Ch BW01 is equal to bandwidth Ch BW02. Each of bandwidth parts BWP02, BWP032, and BWP05 also includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part BWP02 includes data associated with smart RF repeater **612**, bandwidth part BWP032 includes data associated with smart RF repeater **614**, and bandwidth part BWP05 includes data associated with wireless communication node **606**.

Smart RF repeater **610** receives a message associating bandwidth parts of wireless communication beam **616** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part BWP01, which is destined for smart RF repeater **610**. The message may associate, for example, (a) bandwidth part BWP031 with smart RF repeater **614** and (b) bandwidth part BWP04 with wireless communication node **604**. In some embodiments, the message is radio resource control (RRC) message. Smart RF repeater **610** then forwards data associated with each of bandwidth parts BWP031 and BWP04 to its intended destination according to destinations specified in the received message. For example, smart RF repeater **610** may receive wireless communication beam **616** in a first time slot, and smart RF repeater **610** may then forward data associated with bandwidth part BWP04 to wireless communication node **604** in a second time slot via a wireless communication beam **620**, and smart RF repeater **610** may forward data associated with bandwidth part

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BWP031 to smart RF repeater **614** in a third time slot via a wireless communication beam **622**.

Smart RF repeater **612** receives a message associating bandwidth parts of wireless communication beam **618** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part BWP02, which is destined for smart RF repeater **612**. The message may associate, for example, (a) bandwidth part BWP032 with smart RF repeater **614** and (b) bandwidth part BWP05 with wireless communication node **606**. In some embodiments, the message is a RRC message. Smart RF repeater **612** then forwards data associated with each of bandwidth part BWP032 and bandwidth part BWP05 to its intended destination according to destinations specified in the received message. For example, smart RF repeater **612** may receive wireless communication beam **618** in a fourth time slot, and smart RF repeater **612** may then forward data associated with bandwidth part BWP05 to wireless communication node **606** in a fifth time slot via a wireless communication beam **624**, and smart RF repeater **612** may route data associated with bandwidth part BWP032 to smart RF repeater **614** in a sixth time slot via a wireless communication beam **626**.

Smart RF repeater **614** aggregates (a) data associated with bandwidth part BWP031 received via wireless communication beam **622**, and (b) data associated with bandwidth part BWP032 received via wireless communication beam **626**, to yield aggregated data associated with bandwidth part BWP03, as illustrated in FIG. 6. In some embodiments, bandwidth part BWP03 has a bandwidth Ch BW03 which is greater than either of bandwidth Ch BW01 or Ch BW02. In some embodiments, smart RF repeater **614** receives each of wireless communication beams **622** and **626** in different respective time slots. Smart repeater **614** then forwards aggregated data associated with bandwidth part BWP03 to wireless communication node **608** via a wireless communication beam **628**. However, in some alternate embodiments, smart RF repeater **614** is the final destination of aggregated data associated with bandwidth part BWP03. Additionally, in some alternate embodiments, smart repeater **614** is replaced with another device, such as CPE, a UE device, or a wireless base station, that is capable of aggregating data associated with of bandwidth part BWP031 and bandwidth part BWP032 to yield aggregated data associated with bandwidth part BWP03.

In some embodiments, each smart RF repeater **610**, **612**, and **614** forwards received data that is destined for another wireless communication node without processing the data. Accordingly, in particular embodiments, error management messages, e.g., error correction messages and/or error control messages, are transported from an origination wireless communication node to a destination wireless communication node via one or more smart RF repeaters **610**, **612**, **614** without processing the error correction messages, thereby achieving end-to-end error correction and/or error control. Examples of such error correction messages include, but are not limited to, hybrid automatic repeat request (HARQ) messages, such as HARQ ACK messages and/or HARQ NACK messages. Additionally, some embodiments of wireless communication network **600** are configured to retransmit a data structure, such as a transport block (TB), via an alternate path in the event that the data structure, as received by the destination node, includes one or more errors.

For instance, FIG. 7 is a block diagram of a wireless communication network **700**, which is one embodiment of wireless communication network **600** configured to retransmit a data structure via an alternate path in response to the

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data structure including errors. Wireless communication nodes 604, 606, and 608 of FIG. 6 are not shown in FIG. 7 for illustrative clarity. FIG. 7 depicts an example where a transport block included in bandwidth part BWP031 includes an error. In particular, wireless communication node 602 transmits a wireless communication beam 702 to smart RF repeater 610, where wireless communication beam 702 includes bandwidth parts BWP01, BWP031, and BWP04. Smart RF repeater 610 then forwards data associated with bandwidth part BWP031 to smart RF repeater 614 via a wireless communication beam 704. Bandwidth part BWP031 includes a plurality of transport blocks, although only one transport block, i.e., transport block TBx01, is shown in FIG. 7 for illustrative clarity. Each transport block includes a plurality of coding block groups (CBGs) (not shown), and each coding block group includes a plurality of coding blocks (CBs) (not shown).

Smart RF repeater 614 processes each coding block of each coding block group in each transport block, to determine if there are any errors. Additionally, smart RF repeater 614 sends a HARQ ACK message to wireless communication node 602 via smart RF repeater 610 and/or 612 in response to each data unit that is successfully received by smart repeater 614, and smart RF repeater 614 sends a HARQ NACK message to wireless communication node 602 via smart RF repeater 610 and/or 612 in response to each data unit that is not successfully received by smart repeater 614. As such, wireless communication network 700 supports end-to-end HARQ between wireless communication nodes 602 and 604 and across one or more smart RF repeaters.

In the FIG. 7 example, smart repeater 614 determines that one coding block in a first copy 705 of transport block TBx01 includes an error. Specifically, coding block CB-031-x02 includes an error as symbolically shown in FIG. 7 by the coding block being filled with cross-hatching, while coding blocks CB-031-x01 and CB-031-x03 do not include errors, as symbolically shown in FIG. 7 by the coding block being filled with light shading. Smart repeater 614 then sends a message to wireless communication node 602 advising of the error in transport block TBx01, within a prescribed time interval associated with coding block CB-031-x02.

Wireless communication node 602 retransmits transport block TBx01 to smart RF repeater 614 via a different path than that used to originally send transport block TBx01 to smart RF repeater 614, in response to receiving the message indicating the error in transport block TBx01. In particular, wireless communication node 602 transmits a wireless communication beam 706 to smart RF repeater 612, with transport block TBx01 appended to the end of bandwidth part BWP0321. Smart RF repeater 612 then forwards data associated with BWP0321 to smart RF repeater 614 via a wireless communication beam 704, such that smart RF repeater 614 receives a second copy 710 of transport block TBx01 via bandwidth part BWP0321.

Coding blocks CB-031-x02 and CB-031-x03 in second copy 710 of transport block TBx01 are error free. However, coding block CB-031-x01 in second copy 710 of transport block TBx01 includes an error. Nevertheless, smart RF repeater 614 has received error-free copies of each coding blocks CB-031-x01, CB-031-x02, and CB-031-x03, between first copy 705 and second copy 710 of transport block TBx01. Accordingly, smart RF repeater 614 generates a combined copy 712 of transport block TBx01 from error-free coding blocks of each of first copy 705 of TBx01 and second copy of TBx01, using, for example, a chase combining process. While the FIG. 7 example illustrates com-

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bined copy 712 transport block of TBx01 being formed from only two copies of the transport block, combined copy 712 of transport block of TBx01 could be formed using additional copies of the transport block, such as by executing a chase combining process a plurality of times.

Combinations of Features

Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations.

(A1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths.

(A2) The method denoted as (A1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the selected one of the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(A3) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(A4) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(A5) In any one of the methods denoted as (A1) through (A4), the measured conditions of the plurality of wireless communication paths may include one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

(A6) In any one of the methods denoted as (A1) through (A5), selecting one of the plurality of wireless communication paths to use for transferring data between the first node of the wireless communication network and the second node of the wireless communication network may include selecting one or more wireless communication beams of the selected one of the plurality of wireless communication paths.

(B1) A method for multiple repeater aggregation includes (1) receiving a plurality of copies of a wireless communication signal, each copy of the wireless communication signal being received via a respective wireless communica-

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tion path of a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) combining the plurality of copies of the wireless communication signal to generate a combined wireless communication signal.

(B2) In the method denoted as (B1), combining the plurality of copies of the wireless communication signal to generate the combined wireless communication signal may include soft combining the plurality of copies of the wireless communication signal.

(B3) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a user equipment device.

(B4) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a wireless base station.

(B5) In either one of the methods denoted as (B1) through (B4), the wireless communication signal may include one or more of downlink data and uplink data.

(B6) Any one of the methods denoted as (B1) through (B5) may further include receiving each copy of the wireless communication signal via a respective wireless communication beam.

(C1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, a plurality of wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication signals representing different data, and each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) aggregating the received plurality of wireless communication signals at the first node.

(C2) The method denoted as (C1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(C3) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(C4) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(C5) In any one of the methods denoted as (C1) through (C4), at least two wireless communication signals of the plurality of wireless communication signals may be part of a common wireless communication beam.

(C6) In any one of the methods denoted as (C1) through (C5), at least two wireless communication signals of the plurality of wireless communication signals may be within different respective frequencies ranges and/or may be within different respective time ranges.

(D1) A method for load balancing, includes receiving, from a donor wireless communication node, a message, the message associating (a) a first bandwidth part with a first destination wireless communication node, and (b) a second bandwidth part with a second destination wireless communication node.

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(D2) The method denoted as (D1) may further include (1) receiving, in a first time slot, data associated with the first bandwidth part, and (2) forwarding the data associated with the first bandwidth part to the first destination wireless communication node, in a second time slot.

(D3) The method denoted as (D2) may further include (1) receiving, in the first time slot, data associated with the second bandwidth part, and (2) forwarding the data associated with the second bandwidth part to the second destination wireless communication node, in a third time slot.

(D4) In any one of the methods denoted as (D1) through (D3), the donor wireless communication node may be a wireless base station.

(E1) A method for load balancing includes (1) receiving, via a first wireless communication beam, data associated with a first bandwidth part of the first wireless communication beam, (2) receiving, via a second wireless communication beam, data associated with a first bandwidth part of the second wireless communication beam, and (3) aggregating (a) the data associated with the first bandwidth part of the first wireless communication beam and (b) the data associated with the first bandwidth part of the second wireless communication beam.

(E2) The method denoted as (E1) may further include receiving the data associated with the first bandwidth part of the first wireless communication beam and the data associated with the first bandwidth part of the second wireless communication beam in different respective time slots.

(E3) Either one of the methods denoted as (E1) and (E2) may further include (1) receiving the data associated with the first bandwidth part of the first wireless communication beam from a first radio frequency (RF) repeater, and (2) receiving the data associated with the first bandwidth part of the second wireless communication beam from a second radio frequency (RF) repeater.

(F1) A method for end-to-end error management in a wireless communication network includes (1) receiving, at a first wireless communication node, data from a second wireless communication node via a first radio-frequency (RF) repeater, and (2) sending one or more error management messages associated with the data from the first wireless communication node to the second wireless communication node via one or more RF repeaters.

(F2) In the method denoted as (F1), the one or more error management messages may include one or more hybrid automatic repeat request (HARQ) messages.

(F3) In either one of the methods denoted as (F1) and (F2), the data received via the first RF repeater may include one or more errors, and the method may further include receiving a copy of the data via a second RF repeater.

(F4) The method denoted as (F3) may further include generating a combined copy of the data from (a) the data received via the first RF repeater and (b) the copy of the data received via the second RF repeater.

(F5) In the method denoted as (F4), generating the combined copy of the data may include using a chase combining process.

Changes may be made in the above methods, devices, and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

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What is claimed is:

1. A method for multiple repeater aggregation, comprising:

receiving, at a first node of a wireless communication network,
 wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths,
 each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater;
 and
 selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths,
 wherein the measured conditions of the plurality of wireless communication paths include one or more of
 a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths,
 a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths,
 a time advance of each wireless communication path of the plurality of wireless communication paths,
 a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths,
 a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and
 a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

2. The method of claim 1, further comprising causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the selected one of the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

3. The method of claim 1, wherein:

the first node of the wireless communication network includes a user equipment (UE) device; and
 the second node of the wireless communication network includes a wireless base station.

4. The method of claim 1, wherein:

the first node of the wireless communication network includes a wireless base station; and
 the second node of the wireless communication network includes a user equipment (UE) device.

5. The method of claim 1, wherein selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network comprises selecting one or more wireless communication beams of the selected one of the plurality of wireless communication paths.

6. A method for multiple repeater aggregation, comprising:

receiving a plurality of copies of a wireless communication signal,
 each copy of the wireless communication signal being received via a respective wireless communication path of a plurality of wireless communication paths,

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each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater;
 combining the plurality of copies of the wireless communication signal to generate a combined wireless communication signal; and
 selecting one of the plurality of wireless communication paths for use in transferring additional data between a first node of a wireless communication network and a second node of the wireless communication network,

at least partially based on measured conditions of the plurality of wireless communication paths,
 wherein the measured conditions of the plurality of wireless communication paths include one or more of
 a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths,
 a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths,

a time advance of each wireless communication path of the plurality of wireless communication paths,
 a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths,
 a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and
 a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

7. The method of claim 6, wherein combining the plurality of copies of the wireless communication signal to generate the combined wireless communication signal comprises soft combining the plurality of copies of the wireless communication signal.

8. The method of claim 6, wherein the steps of receiving and combining are performed at a user equipment device.

9. The method of claim 6, wherein the steps of receiving and combining are performed at a wireless base station.

10. The method of claim 6, wherein the wireless communication signal includes one or more of downlink data and uplink data.

11. The method of claim 6, further comprising receiving each copy of the wireless communication signal via a respective wireless communication beam.

12. A method for multiple repeater aggregation, comprising:

receiving, at a first node of a wireless communication network,
 a plurality of wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths,

each of the plurality of wireless communication signals representing different data, and
 each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater;
 aggregating the received plurality of wireless communication signals at the first node; and
 selecting one of the plurality of wireless communication paths for use in transferring additional data between the first node of the wireless communication network and the second node of the wireless communication network,

at least partially based on measured conditions of the plurality of wireless communication paths,
 wherein the measured conditions of the plurality of wireless communication paths include one or more of

a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths,
 a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths,

a time advance of each wireless communication path of the plurality of wireless communication paths,
 a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths,
 a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and
 a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

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a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths,

a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths,

a time advance of each wireless communication path of the plurality of wireless communication paths,

a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths,

a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and

a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

13. The method of claim **12**, further comprising causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the plurality of wireless com-

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munication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

14. The method of claim **12**, wherein:

the first node of the wireless communication network includes a user equipment (UE) device; and

the second node of the wireless communication network includes a wireless base station.

15. The method of claim **12**, wherein:

the first node of the wireless communication network includes a wireless base station; and

the second node of the wireless communication network includes a user equipment (UE) device.

16. The method of claim **12**, wherein at least two wireless communication signals of the plurality of wireless communication signals are part of a common wireless communication beam.

17. The method of claim **12**, wherein at least two wireless communication signals of the plurality of wireless communication signals are within different respective frequencies ranges and/or are within different respective time ranges.

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